AIIDE 2026 POSTER · PROCEDURAL CONTENT GENERATION

Did You Get the Level You Asked For?

Bridging LLMs and PCGRL for *Super Mario Bros.* Level Generation

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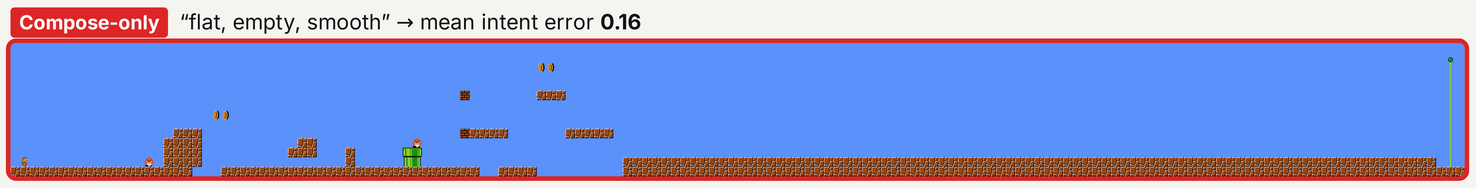


! The problem: generators are material-bound

Designers think in intent: **"flat, empty, smooth."** A corpus-trained generator still hands back a generic Mario level with enemies, gaps and bumps.

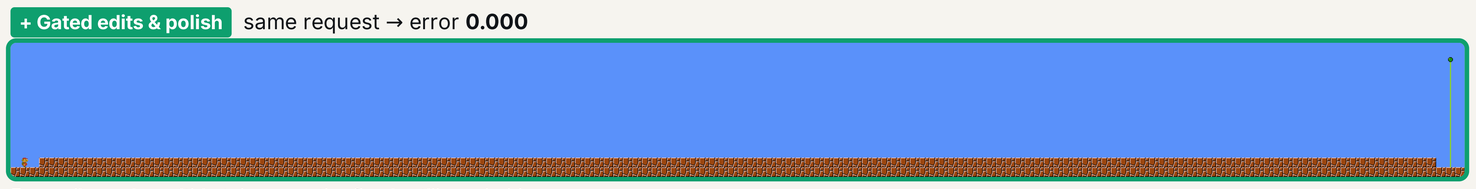
If every corpus pattern contains an enemy, **no recombination of corpus patterns contains none**. In our corpus, the 5th percentile of enemy density is 0.08 and of gap difficulty 0.11. The request lies outside what composition can express.

Compose-only "flat, empty, smooth" → mean intent error **0.16**



Corpus material forces enemies, gaps, coins and relief.

+ Gated edits & polish same request → error **0.000**

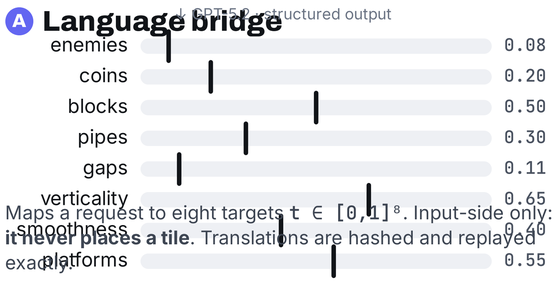


Every dimension within tolerance; the flag is still reachable.

1 Approach: separate what was asked from how it is built

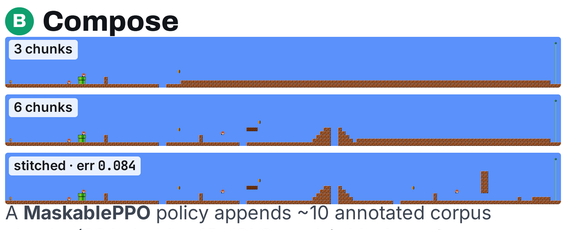
"A peaceful stroll: no enemies, no gaps, just a few coins."

A Language bridge — LoRA T5-Base, structured output



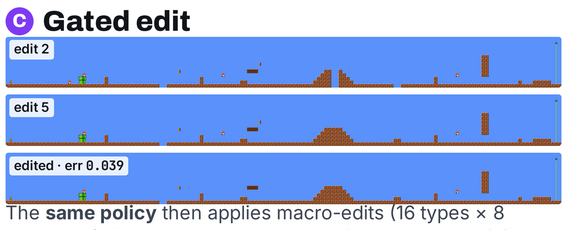
Maps a request to eight targets $t \in [0,1]^8$. Input-side only: **it never places a tile**. Translations are hashed and replayed exactly.

B Compose



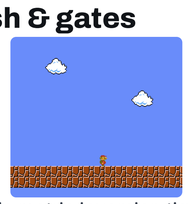
A **MaskablePPO** policy appends ~10 annotated corpus chunks (264 chunks, 15 VGLC levels). Masks enforce seam physics, family order and width.

C Gated edit



The **same policy** then applies macro-edits (16 types × 8 segments). Each edit is validated and flood-checked; **failures are reverted**.

D Polish & gates



- ✓ structure
- ✓ reachability
- ✓ A* agent wins

Deterministic metric inversion tightens residuals (tol. 0.05). Release requires an external **A* agent** to finish the level.

One terminal reward, γ = 1, for both phases: $R = 4 \cdot align + 2 \cdot style + 3 \cdot play - dup$

align: mean + worst-dimension error · *style*: e^{-TPKL} vs. corpus 3×3 patterns · *play*: spawn-to-flag flood · *dup*: chunk repeats

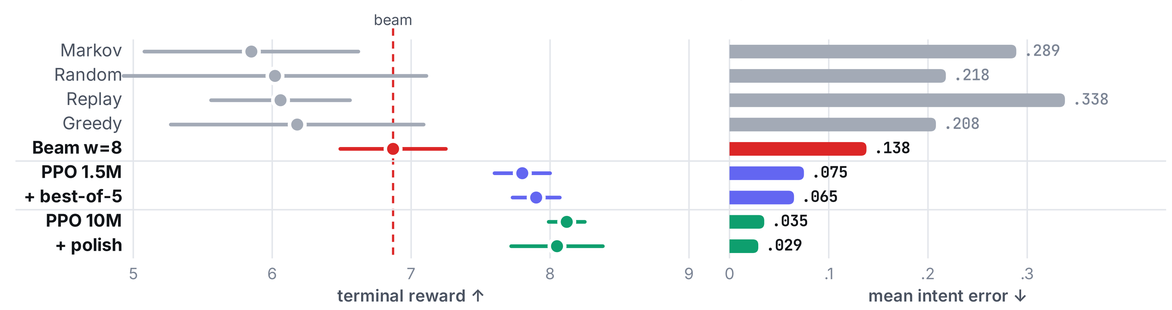
2 An eight-number design contract

- **Enemies** — goomba density
- **Coins** — coin density
- **Blocks** — bricks and ? blocks
- **Pipes** — pipe tile density
- **Gaps** — just-impossible gap equivalents
- **Verticality** — amplitude of the path envelope
- **Smoothness** — 1 − mean envelope step
- **Platforms** — path share above ground

Achievable-band re-normalization. Structural metrics are rescaled to the 5th–95th percentile of what legal compositions can reach, so 0 means *as little as the medium permits*, 1 *as much as it permits*. Platform-usage translation error fell from 0.137 to 0.053.

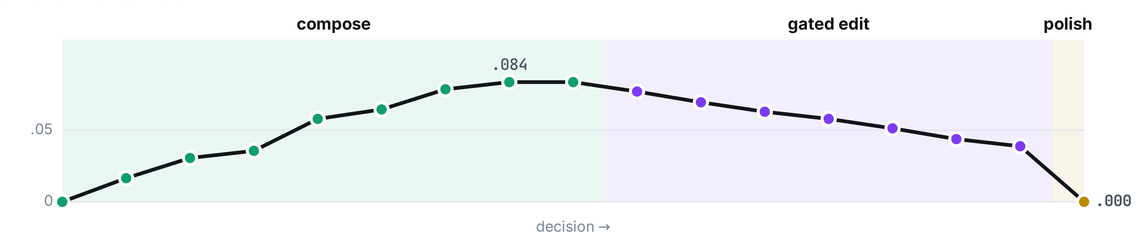
RQ1 Learning beats search

+1.0 reward over beam search, all 3 seeds ($p = 10^{-4}$) **7×** faster: 64 ms vs 0.47 s per level



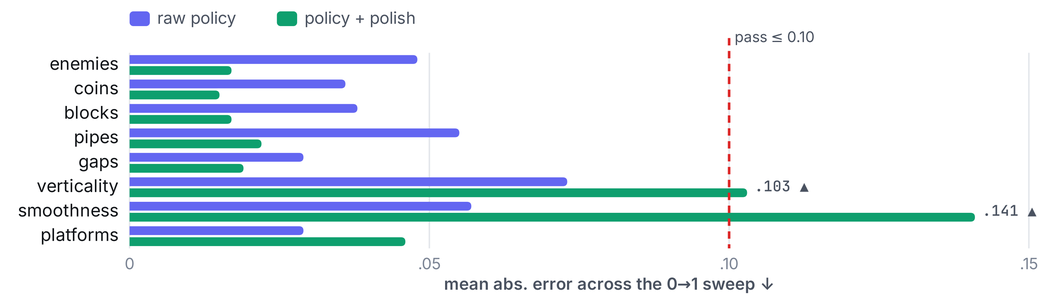
3 One episode, two phases

Target **"flat, empty, smooth"** (final policy). Corpus chunks push the error up; gated edits pull it down; polish closes the residual.



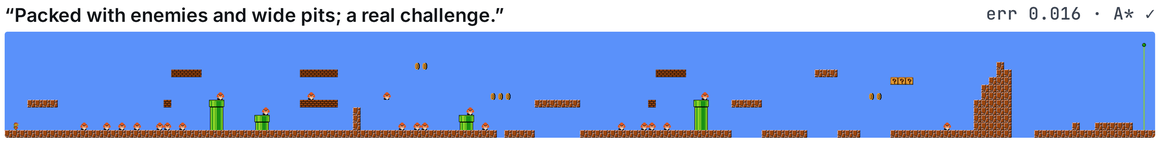
RQ2 Control reaches the boundaries

8/8 sweeps pass (MAE ≤ 0.10, ρ ≥ 0.8), incl. exact-zero enemies and gaps. ▲ = residual reported, not hidden

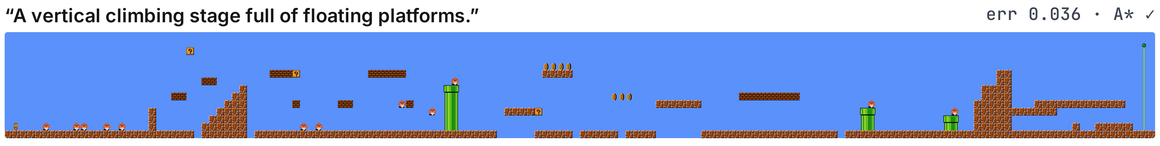


4 Ask, and get

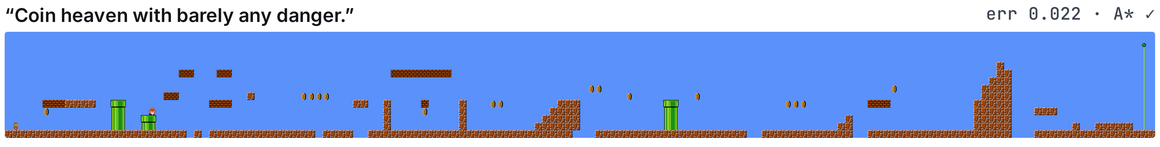
"Packed with enemies and wide pits; a real challenge." err 0.016 · A* ✓



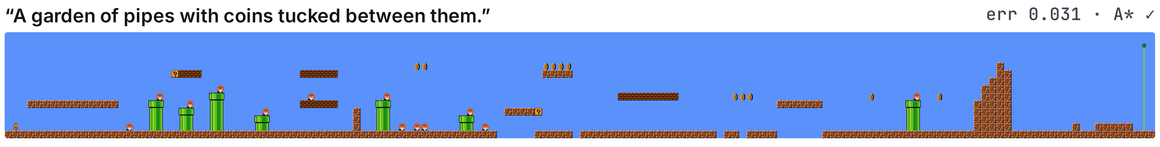
"A vertical climbing stage full of floating platforms." err 0.036 · A* ✓



"Coin heaven with barely any danger." err 0.022 · A* ✓

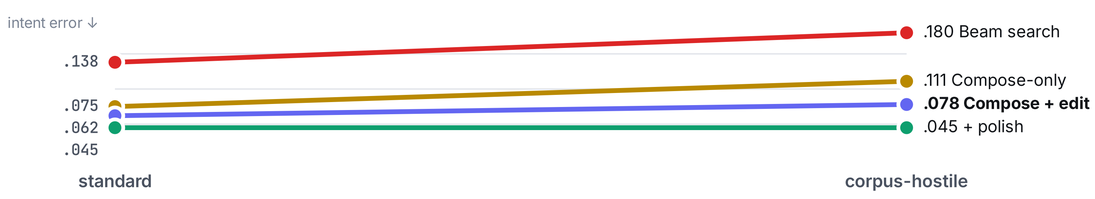


"A garden of pipes with coins tucked between them." err 0.031 · A* ✓



RQ3 Editing matters off-corpus

−30% error vs. compose-only on corpus-hostile targets ($p = 0.0095$)



RQ4 Language is a monotonic interface

0.053 translation MAE [0.042, 0.067] over 46 descriptions



Translation is now the larger error term: end-to-end error 0.035 → 0.081.

Takeaway

The request becomes a **numeric contract**. When an output is wrong, the trace says whether the **translation** missed, the **generator** missed, or the targets **conflict**.

Scope. Bounded structural control over a fixed 8-dim vocabulary; one game, 15 VGLC overworld levels; no designer study; A* completion ≠ human difficulty.

Funded in part by CAPES, CNPq and Fapemig.

Interactive slides & demo

Code, models & data